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MILITARY

Chang Hong-4 (CH-4) Rainbow-4

The CH-4 is built by state-owned aircraft manufacturer China Aerospace Science and Technology Corporation (CASC) and closely resembles the U.S.-made General Atomics Aeronautical Systems

MQ-9 Reaper drone. At Airshow China in Zhuhai in 2014, a company brochure said the CH-4 series is suitable for high-altitude missions over land and sea, and has a maximum take-off weight of 1,260kg (2,770lb) and payload capacity of 115kg. Endurance for the CH-4 series is claimed to be up to 30h.

The CH-series unmanned aircraft have been sold to armed forces in more than 10 countries in the Middle East, Southeast Asia and Africa, making them the largest military drone family China has exported, according to Qin Yongming, general manager of the academy's unmanned aircraft business. The most popular model to date-the CH-4-is well regarded by its users for its excellent service records in several regional wars and conflicts. The drone had conducted more than 4,000 flights over the past decade up to 2021 under the control of its foreign operators, boasting over 20,000 flying hours and an astonishing kill rate of 99 percent in its more than 800 missile strikes, he said.

The China Academy of Aerospace Aerodynamics is a Beijing-based subsidiary of China Aerospace Science and Technology Corp, a State-owned defense conglomerate. It started to research and develop the CH drones in 2000.

The CH-4, has a wingspan of 18m, a take-off weight of 1.3 tons and a payload of 350kg. The Chinese UAV can carry a payload including Lan Jian 7 (Blue Arrow 7) laser-guided air-to-surface missiles, TG100 laser/INS/GPS-guided bombs, and AR-1/HJ-10 anti-tank missile the Chinese equivalent to the American-made Hellfire missile.

It is specifically designed for high-altitude missions over land and sea, and can fire its weapon from up to 5,000 meters. The CH-4 also has a retractable electro-optical sensor turret, and a datalink back to the ground control station. The CH-4 also boasts a modern, two person control station to fly the drone remotely, with provisions for both line of sight and satellite communications, Popular Science notes. The video clip also shows what appears to be Chinese support crew in the ground control station.

Franz-Stefan Gady reported October 16, 2015 that the CH-4B Rainbow unmanned aerial vehicle (UAV) was shown in an Iraqi video took off for its first combat mission against ISIS militants in Anbar province. The short video clip also features footage of the drone firing at an alleged ISIS target with an AR-1/HJ-10 anti-tank missile.

Iraq had received an unknown number of CH-4B combat drones in early 2015. It is not clear whether China supplied Iraq with just the CH-4B mixed attack and reconnaissance UAV shown in the video, or also with the CH-4A variant, a reconnaissance drone with a 5,000km range and 30-40 hour endurance.



The Saudi official body announced that China has purchased the Rainbow 4 unmanned aerial vehicle production plan to achieve a powerful domestic UAV. Although this news has been widely reported, this is the official body for the first time to confirm the existence of this contract.

According to Jane's magazine, Saudi Arabia's most critical scientific and technological development agency, "Abdullah Aziz King Technology City and Education and Training" (KACST) announced that on 16 March 2017 and China Aerospace Science and Technology Group Company (CASC) signed a "partner" document, together to achieve the manufacture of rainbow-4 in Saudi Arabia hit one unmanned aerial vehicle.

During the visit to Saudi Arabia, Saudi Arabia and China signed a total of 65 billion US dollars of various types of contracts. According to reports, in February, the two sides specifically discuss the rainbow-4 manufacturing cooperation agency Saudi Technology Development Investment Corporation (TAQNIA) and China Aerospace Long March International Trade Co., Ltd. (ALIT), has reached a basic agreement. Jane's Weekly reported that Saudi Rainbow-4 will be the third overseas plant to produce this type of UAV, the first two in Pakistan and Myanmar. But the specific of these factories were able to bear the depth of the manufacturing work, is still unknown. It is expected that the Saudi plant will also be responsible for the maintenance and overhaul of Saudi related unmanned aerial vehicles.

Rainbow -4 has been exported to Egypt, Saudi Arabia, Iraq, Jordan and other Middle East countries. Its carrying the most important weapons AR-1 air missiles, hit with an accuracy of 1.5 meters or so, can effectively destroy mobile vehicles or solid buildings.

The US government took a very conservative attitude for the export of highly controversial combat UAV also limited and binding position, which makes Saudi Arabia and other countries feel very difficult to access MQ-1 predators and other advanced unmanned aerial vehicles. Developed MQ-1 General Atomic Company even in the congressional hearing that their business is seriously affected by the relevant policies, can not provide to the Middle East customers developed UAV.

In any case, Rainbow-4 unmanned aerial vehicle production line of the overall export, proved that the advanced performance of this advanced weapon. China is no longer export cheap low-grade weapons and equipment of the country, will be in the international military market.

In 2018 the Trump Administration announced the launch of an assessment of the UAV export policy during the Obama administration. It is speculated that the United States was likely to relax the UAV export restrictions to compete with China in the international arms market. Faced with the challenges of the future, China's main export of drones "Rainbow" and "Pterosaur" two series also opened the upgrade mode.

The Chinese Academy of Aerospace Group's upgraded version of "Rainbow-4" has completed the first intensive bombing test in extreme environments. According to reports, in January 2018, the upgraded version of the "Rainbow-4" drone took six flight days in the northwestern China test base, braving the extreme darkness and other extreme environments such as the morning sun, and multi-type ammunition. Carrying verification was carried out. Usually the natural environment such as fog, moderate rain and light snow may affect some of the guided weapons, but because of the test of the "Rainbow-4" drone with multi-point and hybrid weapon mounting capabilities, can be different The weather adopts different forms of guidance to achieve a precise impact on the target.

This overcame the shortcomings of previous single weapons and laser-guided weapons. Among them, the 50 kg class mother bomb can effectively implement the regional blockade, killing the enemy's living forces and airport fortifications and other hard and soft targets; the 50 kg class mother-end bomb can cause a devastating blow to the armored targets such as enemy tanks, infrared The model can track the thermal dynamics in time, the millimeter wave type can capture the target with super precision; the 50 kg class satellite guided bomb relies on satellite navigation to safely and secretly launch a fatal blow in the higher airspace; the 100 kg laser guided bomb gives no The man-machine "the first level of the enemy in the army" is capable; the 100-kilometer satellite-guided glide bomb can launch a surprise attack more than ten kilometers away.

The "Rainbow" drone team said that the drone tested this time is "Rainbow-4 Enhanced", which has better load capacity and power supply capacity than the earlier models, and a new in-flight data stream. Architecture. These new technologies will also be used in the new generation of "Rainbow-4C" drones. The results of the live-fire target test fully met the design specifications, and all the ammunition models participating in the test met the standards.

The multi-level and multi-type weapons and ammunition carrying test of the "Rainbow-4" drone successfully laid a solid foundation for the great enrichment of the ammunition maps of the future series models. The dual advantages of multi-hanger with multiple bombs means that the Rainbow series of drones can effectively attack more targets at a higher distance in the future. Combined with the advantages of a variety of guide heads, whether it is a soft or hard target or a point target, it can achieve effective killing, so that the "Rainbow" series of drones will continue to evolve with the complex battlefield environment.

drones in the international arms market. In stark contrast, Chinese drones continue to attack the overseas market, whether it is the official video released by Iraq, the parade of Pakistan and Turkmenistan, or the promotional video of Nigeria in 2017 and the Twitter of the Saudi Air Force.

The British "Jane's Defense Weekly" website revealed that the traditional American allies such as the United Arab Emirates and Pakistan introduced the Chinese "Pterosaur" series of drones. The Trump administration was seeking to ease the restrictions on drone exports in an attempt to regain its dominance in the market. To this end, China is also constantly pushing for technological upgrading.

According to the "Global Times" reporter, the next "Rainbow" drone team will also conduct a joint flight test of "Rainbow-3" and "Rainbow-4" drones to achieve formation operations between different models of UAVs. Information sharing. The new generation of "Pterosaur-1D" drones with carbon fiber composite fuselage will also be sold in 2018.

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